Informatica® Cloud Data Integration

SAP OData V2 Connector

Informatica Cloud Data Integration SAP OData V2 Connector
April 2026

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Preface

Use SAP OData V2 Connector to learn how to read from or write to OData V2-compliant applications in SAP by using Data Integration. Learn to create an SAP OData V2 connection, develop and run mappings, and mapping tasks in Data Integration.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

IDMC web site

You can access the Informatica® Intelligent Data Management Cloud™ (IDMC) web site at http://www.informatica.com/cloud.

IDMC Communities

Use the IDMC Community to discuss and resolve technical issues. You can also find technical tips, documentation updates, and answers to frequently asked questions.

Access the IDMC Community at:

https://network.informatica.com/community/informatica-network/products/cloud-integration

Developers can learn more and share tips at the Cloud Developer community:

https://network.informatica.com/community/informatica-network/products/cloud-integration/cloud-developers

IDMC Marketplace

Visit the Informatica Marketplace to try and buy Data Integration Connectors, templates, and mapplets:

https://marketplace.informatica.com/

Data Integration connector documentation

You can access documentation for Data Integration Connectors at the Documentation Portal. To explore the Documentation Portal, visit https://docs.informatica.com.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

IDMC Trust Center

The IDMC Trust Center provides information about Informatica security, compliance, and privacy practices.

You can access the trust center at https://trust.informatica.com/.

IDMC Pulse Dashboard

The IDMC Pulse Dashboard provides system status and upgrade information for all PODs and services across IDMC.

Follow the services you use and subscribe to alerts to receive notifications about upgrades, maintenance, production incidents, and upcoming releases. The IDMC Pulse Dashboard displays the production status of all the IDMC products. All maintenance updates are posted to this page, and during an outage, it will have the most current information. To ensure you are notified of updates and outages, you can subscribe to receive updates for a single service or all IDMC services. Subscribing to all services is the best way to be certain you never miss an update.

For more information, see the Service Status and Communications guide.

Informatica Global Customer Support

You can contact a Global Support Center through the Informatica Network or by telephone.

To find online support resources on the Informatica Network, click **Contact Support** in the IDMC Help menu to go to the **Cloud Support** page. The **Cloud Support** page includes system status information and community discussions. Log in to Informatica Network and click **Need Help** to find additional resources and to contact Informatica Global Customer Support through email.

The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at https://www.informatica.com/services-and-training/support-services/contact-us.html.

CHAPTER 1

Introduction to SAP OData V2 Connector

You can use SAP OData V2 Connector to access SAP applications from Data Integration. You can read from or write to OData V2-compliant applications in SAP deployed in the cloud or on-premises.

SAP OData V2 (Open Data Protocol Version 2) is a service that defines best practices to build and consume RESTful APIs. The SAP OData V2 service helps you to create, read, update, and delete SAP data in a standardized way in the data integration process. SAP OData V2 Connector communicates with the SAP OData APIs version 2 that are exposed through the SAP OData V2 service and enables effective data integration between the SAP system and other applications.

Use SAP OData V2 Connector to read data from hierarchical and flat entities, and write data to flat entities.

When you use SAP OData V2 Connector, you can create an SAP OData V2 connection and use the connection both as a source and a target in a mapping and task. When you run the mapping or task, the agent reads data from or writes data to entities that are exposed and accessible through the OData V2-compliant application in SAP based on the workflow and SAP OData V2 connection configuration.

You can switch mappings to advanced mode to include transformations and functions that enable advanced functionality.

Example

You want to read and filter sales order data from SAP S/4HANA. Use SAP OData V2 Connector to call the required sales order OData V2 API in SAP S/4HANA. You can then process the response obtained from SAP S/4HANA.

You can use the SAP OData V2 Connector to communicate with the SAP OData V2 service APIs to read and filter data related to business partner, customer, supplier, and vendor entities. You can also use transformations on data for better results.

SAP OData V2 Connector assets

Create assets in Data Integration to integrate data using SAP OData V2 Connector.

When you use SAP OData V2 Connector, you can include the following Data Integration assets:

- Dynamic mapping task
- Mapping
- Mapping task

For more information about configuring assets and transformations, see *Mappings*, *Transformations*, and *Tasks* in the Data Integration documentation.

CHAPTER 2

Connections for SAP OData V2

Create an SAP OData V2 connection to securely read from or write to OData V2-compliant applications in SAP deployed in the cloud or on-premises.

You can use an SAP OData V2 connection to specify sources and targets in mappings, dynamic mapping tasks, and mapping tasks.

Prerequisites

To connect to SAP OData V2 services, the SAP administrator needs to configure the SAP user account in the SAP system.

Additionally, when you configure the connection properties, you need to complete the authentication prerequisites.

Configure SAP user account

The SAP administrator configures the S_SERVICE authorization object for the user to access OData V2-compliant applications in the SAP system.

The following table lists the fields and values required for the S_SERVICE authorization object to configure user permissions in SAP:

Field	Value
SRV_NAME	Provide SAP service names to which you want to connect.
SRV_TYPE	Select the **Hash Value for TADIR Object** option from list.

For more information about the authorization object and how to use it, see SAP user authorizations.

Prepare for authentication

You can configure basic, API key, authorization code, and client credentials authentication types to access OData V2-compliant applications in SAP.

Before you configure the connection properties, you need to keep the authentication details handy based on the authentication type that you want to use.

Basic

To connect to SAP OData V2 services using basic authentication, you need the SAP account user name and password.

Get the required details from the SAP application to which you want to connect.

For more information about basic authentication in SAP, see Basic authentication in the SAP documentation.

API key

To connect to SAP OData V2 services using API key authentication, you need a unique API key that SAP OData V2 Connector uses to authenticate the API calls made to the SAP OData endpoint.

Get the API key from the SAP application to which you want to connect.

For more information about how to generate and use an API key, see Add API keys to an environment in the SAP documentation.

Authorization code

To connect to SAP OData V2 services using the OAuth 2.0 authorization code, you need the SAP client ID, client secret, authorization token URL, access token URL, and access token.

Before you use OAuth 2.0 authorization code authentication, complete the following tasks:

- Create an authorization integration in SAP, and register the Informatica redirect URL in SAP Integration Suite.
  SAP Integration Suite is an integration platform-as-a-service that enables clients that support OAuth to redirect users to an authorization page and generate access tokens, and optionally, refresh tokens to access SAP.

  Register the following Informatica redirect URL in SAP Integration Suite:

  `https://<Informatica cloud hosting facility for your organization>/ma/proxy/oauthcallback`

  If the access token expires and the response returns 401 error code, Informatica redirect URL, which is outside the customer firewall, tries to connect to the endpoint and retrieves a new access token.

  For more information about how to create an authorization integration and get the authorization details, see OAuth 2.0 authorization code in the SAP documentation.
- Import the `.cer` certificate into the cacerts file located in the jdk directory available at one or more of the following locations within your Secure Agent installation:

  - `<Secure Agent installation directory>/jdk/jre/lib/security`
  - `<Secure Agent installation directory>/jdk/lib/security`
  - `<Secure Agent installation directory>/jdk8/jre/lib/security`

  Ensure to add the certificate to all the available directories.
- If there is a jdk directory within the `<Secure agent installation directory>\apps` folder, navigate to the following directory and import the `.cer` certificate into the `cacerts` file located in the jdk directory available at one or more of the following locations within your Secure Agent installation:

  - `<Secure agent installation directory>\apps\jdk\zulu8<latest_version>\jre\lib\security`
  - `<Secure agent installation directory>\apps\jdk\zulu17<latest_version>\lib\security`

After you import the `.cer` certificate into the cacerts file, you need to restart the Secure Agent.

Client credentials

To connect to SAP OData V2 services using OAuth 2.0 client credentials, you need the SAP client ID, client secret, access token URL, and access token.

Before you use OAuth 2.0 client credentials authentication, complete the following tasks:

- Configure the OAuth endpoint with the client credentials grant type and then create an authorization integration to get the authorization details in the SAP system.
  For more information about how to create an authorization integration and get the authorization details, see OAuth 2.0 client credentials in the SAP documentation.
- Import the `.cer` certificate into the cacerts file located in the jdk directory available at one or more of the following locations within your Secure Agent installation:

  - `<Secure Agent installation directory>/jdk/jre/lib/security`
  - `<Secure Agent installation directory>/jdk/lib/security`
  - `<Secure Agent installation directory>/jdk8/jre/lib/security`

  Ensure to add the certificate to all the available directories.
- If there is a jdk directory within the `<Secure agent installation directory>\apps` folder, navigate to the following directory and import the `.cer` certificate into the `cacerts` file located in the jdk directory available at one or more of the following locations within your Secure Agent installation:

  - `<Secure agent installation directory>\apps\jdk\zulu8<latest_version>\jre\lib\security`
  - `<Secure agent installation directory>\apps\jdk\zulu17<latest_version>\lib\security`

After you import the `.cer` certificate into the cacerts file, you need to restart the Secure Agent.

Connect to SAP OData V2

Let's configure the SAP OData V2 connection properties to connect to SAP OData V2 services, and read from or write to OData V2-compliant applications in SAP.

Before you begin

Before you get started, you'll need to configure the SAP system and get the relevant configuration details from your SAP account based on the authentication type that you want to configure.

Check out "Prerequisites" on page 8 to learn more about the configuration prerequisites.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. Select a Secure Agent, Hosted Agent, serverless, or elastic runtime environment. For more information about runtime environments, see *Runtime Environments* in the Administrator help.
Service Type	The service type of the OData V2 application endpoint to which you want to connect. Select one of the following service types from the list: - Default. Connects to a specific SAP service. - SAP Gateway Catalog. Connects to the SAP catalog service that provides a list of all available services on SAP Gateway. Default is Default.

Service URL	The service URL for the selected service type.
	If you select the **Default** service type, enter the root URL of the service in the following format:
	`http://<Host name of the SAP server>:<Port number>/sap/opu/odata/sap/<Service name>/`
	For example, if you want to connect to the ZALL_DATATYPE_SRV service in SAP when the host name is http://invs15con01.informatica.com and port number is 8081, enter the following service URL:
	`http://<Host name of the SAP server>:<Port number>/sap/opu/odata/sap/ZALL_DATATYPE_SRV/`
	If you select the **SAP Gateway Catalog** service type, enter the service URL in the following format:
	`http://<Host name of the SAP server>:<Port number>/sap/opu/odata/IWFND/<Catalog service name>`
	For example, if you want to connect to the CATALOGSERVICE;v=2 catalog service in SAP when the host name is abcd01con02.example.com and port number is 8081, enter the following service URL:
	`http://abcd01con02.example.com:8081/sap/opu/odata/IWFND/CATALOGSERVICE;v=2`

Authentication types

You can configure basic, API key, authorization code, and client credentials authentication types to access OData V2-compliant applications in SAP.

Select the required authentication method and then configure the authentication-specific parameters.

Basic authentication

Basic authentication is the default type which requires at a minimum your SAP account name and password.

The following table describes the basic connection properties for basic authentication:

Property	Description
Username	The user name to connect to the SAP OData V2 service.
Password	The password to connect to the SAP OData V2 service.

Advanced settings

The following table describes the advanced connection properties for basic authentication:

Property	Description

SAP Custom Query Options	Additional custom queries that you can use when you connect to the SAP OData V2 service.
	You can enter multiple SAP custom queries, separated by an ampersand (&), in the following format:
	`<Custom query1>=<value>&<Custom query2>=<value>&<Custom query3>=<value>....`
	For example, if you want to pass the SAP client number and language code when you connect to the SAP OData V2 service, enter the following custom queries:
	`sap-client=400&sap-language=DE`
	When you add queries, ensure that there is no space before and after the equal sign (=).
	For more information about the list of SAP custom queries that you can configure, see SAP URL parameters in the SAP documentation.

API key authentication

API key authentication requires at a minimum a unique API key from the SAP application.

The following table describes the basic connection properties for API key authentication:

Property	Description
API Key	The unique API key that SAP OData V2 Connector uses to authenticate the API calls made to the SAP OData endpoint.

Advanced settings

The following table describes the advanced connection properties for API key authentication:

Property	Description

SAP Custom Query Options	Additional custom queries that you can use when you connect to the SAP OData V2 service.
	You can enter multiple SAP custom queries, separated by an ampersand (&), in the following format:
	`<Custom query1>=<value>&<Custom query2>=<value>&<Custom query3>=<value>....`
	For example, if you want to pass the SAP client number and language code when you connect to the SAP OData V2 service, enter the following custom queries:
	`sap-client=400&sap-language=DE`
	When you add queries, ensure that there is no space before and after the equal sign (=).
	For more information about the list of SAP custom queries that you can configure, see SAP URL parameters in the SAP documentation.

Authorization code authentication

OAuth 2.0 authorization code authentication requires at a minimum the SAP client ID, client secret, authorization token URL, access token URL, and access token.

The following table describes the basic connection properties for OAuth 2.0 authorization code authentication:

Property	Description

Authorization Token URL	The SAP authorization token endpoint of the OAuth 2.0 authorization server that is used to authorize the user request.
	Enter the authorization token URL in the following format:
	`https://<Host name of the server>:<Port number>/sap/bc/sec/oauth2/authorize?sap-client=<SAP client number>`
	For example, If the host name is abcd01con02.example.com and port number is 44301, enter the following authorization token URL when the SAP client number is 800:
	`https://abcd01con02.example.com:44301/sap/bc/sec/oauth2/authorize?sap-client=800`

Access Token URL	The SAP access token endpoint of the OAuth 2.0 authorization server that is used to exchange the authorization code to get an access token.
	Enter the access token URL in the following format:
	`https://<Host name of the server>:<Port number>/sap/bc/sec/oauth2/token?sap-client=<SAP client number>`
	For example, If the host name is abcd01con02.example.com and port number is 44301, enter the following access token URL when the SAP client number is 800:
	`https://abcd01con02.example.com:44301/sap/bc/sec/oauth2/token?sap-client=800`

Client ID	The client identifier of your application generated when you configure the application for OAuth.
Client Secret	The client secret generated for the client ID.
Access Token	The access token granted by the authorization server to access the SAP data. Enter the populated access token value that you get from the OAuth endpoint, or click **Generate Access Token** to populate the access token value.

Advanced settings

The following table describes the advanced connection properties for OAuth 2.0 authorization code authentication:

Property	Description

Scope	The scope of the access request when the SAP OData V2 endpoint has defined custom scopes.
	You can enter multiple scope attributes, each separated by a space, in the following format:
	`<Scope attribute1> <Scope attribute2> <Scope attribute3>....`
	For example, enter the following scope attributes:
	`ZGWSAMPLE_BASIC_0001 /IWFND/SG_MED_CATALOG_0002 ZAPI_CHARTOFACCOUNTS_SRV_0001`

Access Token Parameters	Additional parameters to use with the access token URL. Define the access token parameters in the following JSON format: `[{"Name":"<Parameter name>","Value":"<Parameter value>"}]` For more information about the access token parameters that you can define, see the SAP documentation.

Authorization Code Parameters	Additional parameters to use with the authorization token URL.
	Define multiple parameters, separated by a comma, in the following JSON format:
	`[{"Name":"<Parameter name>","Value":"<Parameter value>"}, {"Name":"<Parameter name>","Value":"<Parameter value>"}]`
	For example, you can use the following maximum age and state parameters when you connect to the SAP OData V2 service:
	`[{"Name":"max_age","Value":60}, {"Name":"state","Value":"test"}]`
	For more information about the authorization code parameters that you can define, see the SAP documentation.

Client Authentication	The method of sending client authentication details for authorization to connect to the SAP OData V2 service. Select one of the following client authentications from the list: - Send client credentials in body. Sends the client ID and client secret for authorization in the body of the request. - Basic auth header. Sends the client ID and client secret for authorization in the header of the request. Default is **Send client credentials in body**.

Property	Description
Refresh Token	The refresh token value. Enter the populated refresh token value that you get from the OAuth endpoint, or click **Generate AccessToken** to populate the refresh token value. If the access token is not valid or expires, the Secure Agent fetches a new access token with the help of the refresh token. **Note:** If the refresh token expires, provide a valid refresh token or regenerate a new refresh token by clicking **Generate Access Token**.

SAP Custom Query Options	Additional custom queries that you can use when you connect to the SAP OData V2 service.
	You can enter multiple SAP custom queries, separated by an ampersand (&), in the following format:
	`<Custom query1>=<value>&<Custom query2>=<value>&<Custom query3>=<value>....`
	For example, if you want to pass the SAP client number and language code when you connect to the SAP OData V2 service, enter the following custom queries:
	`sap-client=400&sap-language=DE`
	When you add queries, ensure that there is no space before and after the equal sign (=).
	For more information about the list of SAP custom queries that you can configure, see SAP URL parameters in the SAP documentation.

Client credential authentication

OAuth 2.0 client credential authentication requires at a minimum the SAP client ID, client secret, access token URL, and access token.

The following table describes the basic connection properties for OAuth 2.0 client credential authentication:

Property	Description

Access Token URL	The SAP access token endpoint of the OAuth 2.0 authorization server that is used to exchange the authorization code to get an access token.
	Enter the access token URL in the following format:
	`https://<Host name of the server>:<Port number>/sap/bc/sec/oauth2/token?sap-client=<SAP client number>`
	For example, If the host name is abcd01con02.example.com and port number is 44301, enter the following access token URL when the SAP client number is 800:
	`https://abcd01con02.example.com:44301/sap/bc/sec/oauth2/token?sap-client=800`

Client ID	The client identifier of your application generated when you configure the application for OAuth.

Property	Description
Client Secret	The client secret generated for the client ID.
Access Token	The access token granted by the authorization server to access the SAP data. Enter the populated access token value that you get from the OAuth endpoint, or click **Generate Access Token** to populate the access token value.

Advanced settings

The following table describes the advanced connection properties for OAuth 2.0 client credential authentication:

Property	Description

Scope	The scope of the access request when the SAP OData V2 endpoint has defined custom scopes.
	You can enter multiple scope attributes, each separated by a space, in the following format:
	`<Scope attribute1> <Scope attribute2> <Scope attribute3>....`
	For example, enter the following scope attributes:
	`ZGWSAMPLE_BASIC_0001 /IWFND/SG_MED_CATALOG_0002 ZAPI_CHARTOFACCOUNTS_SRV_0001`

Access Token Parameters	Additional parameters to use with the access token URL. Define the access token parameters in the following JSON format: `[{"Name":"<Parameter name>","Value":"<Parameter value>"}]` For more information about the access token parameters that you can define, see the SAP documentation.

Property	Description
Client Authentication	The method of sending client authentication details for authorization to connect to the SAP OData V2 service. Select one of the following client authentications from the list: - Send client credentials in body. Sends the client ID and client secret for authorization in the body of the request. - Basic auth header. Sends the client ID and client secret for authorization in the header of the request. Default is **Send client credentials in body**.

SAP Custom Query Options	Additional custom queries that you can use when you connect to the SAP OData V2 service.
	You can enter multiple SAP custom queries, separated by an ampersand (&), in the following format:
	`<Custom query1>=<value>&<Custom query2>=<value>&<Custom query3>=<value>....`
	For example, if you want to pass the SAP client number and language code when you connect to the SAP OData V2 service, enter the following custom queries:
	`sap-client=400&sap-language=DE`
	When you add queries, ensure that there is no space before and after the equal sign (=).
	For more information about the list of SAP custom queries that you can configure, see SAP URL parameters in the SAP documentation.

Configure the load balancing connection

You can use basic authentication type in the SAP OData V2 connection to create a load balancing connection when you connect to the SAP server.

When you configure a connection for load balancing, you can connect to the SAP server with the least load at run time.

Before you configure an SAP OData V2 connection that uses basic authentication type for a load balancing connection, you need to perform the prerequisite tasks.

1. Import the `.cer` certificate into the cacerts file located in the jdk directory available at one or more of the following locations within your Secure Agent installation:
   - `<Secure Agent installation directory>/jdk/jre/lib/security`
   - `<Secure Agent installation directory>/jdk/lib/security`
   - `<Secure Agent installation directory>/jdk8/jre/lib/security`

   Ensure to add the certificate to all the available directories.
2. If there is a jdk directory within the `<Secure agent installation directory>\apps` folder, navigate to the following directory and import the `.cer` certificate into the `cacerts` file located in the jdk directory available at one or more of the following locations within your Secure Agent installation:
   - `<Secure agent installation directory>\apps\jdk\zulu8<latest_version>\jre\lib\security`

- <Secure agent installation directory>\apps\jdk\zulu17<latest_version>\lib\security

3. Restart the Secure Agent.

Proxy server settings

If your organization uses an outgoing proxy server to connect to the Internet, the Secure Agent connects to Informatica® Intelligent Data Management Cloud™ (IDMC) through the proxy server.

You can configure the Secure Agent to use an unauthenticated proxy server on Windows and Linux.

The proxy settings apply to connections used in mappings and in mappings in advanced mode.

To configure the proxy settings for the Secure Agent, configure the Secure Agent through the Secure Agent Manager on Windows or shell command on Linux.

For instructions, see "Configure the proxy settings on Windows" or "Configure the proxy settings on Linux" in *Getting Started* in the Data Integration help.

CHAPTER 3

Mappings and mapping tasks with SAP OData V2 Connector

You can configure a mapping to read from or write to OData V2-compliant applications in SAP.

You can configure a Source or Target transformation to represent an SAP OData V2 object. Add the Source and Target transformations in the mapping canvas and configure the SAP OData V2 source and target properties accordingly. Use the Data Integration Mapping Designer to define and configure a mapping for SAP OData V2 sources and targets.

In advanced mode, the Mapping Designer updates the mapping canvas to include transformations and functions that enable advanced functionality.

SAP OData V2 sources in mappings

To read data from entities that are exposed and accessible through OData V2-compliant applications in SAP, configure an SAP OData V2 object as a source in a mapping.

Specify the name and description of the SAP OData V2 source. Configure the source, query options, and advanced properties for the source object.

The following table describes the source properties that you can configure for an SAP OData V2 source:

Property	Description
Connection	Name of the source connection. You can select an existing connection or create a new connection for the source connection property.
Source Type	Type of the source object. Select **Single Object** as the source type. Default is Single Object.
Object	Name of the SAP OData V2 source object based on the source type selected. Select the source object for a single source. You can also select an existing object from the list.

The following table describes the query options that you can configure for an SAP OData V2 source:

Property	Description
Filter	Filters record based on the filter condition in a read operation. You can configure a simple filter or an advanced filter to filter rows from the source based on a filter condition. You can improve efficiency by filtering early in the data flow. Click **Configure** to add conditions to filter records from the data that the Secure Agent reads from the source. Select one of the following filter options from the list: - Not parameterized. Uses a basic filter to specify the object, field, operator, and value to select specific records. This filter option applies only to flat entities. - Advanced. Uses an advanced filter to define a complex filter condition. This filter option applies to flat and hierarchical entities. If you configure a filter here, do not enter the $filter query in the **Query Data Options** field in the advanced source properties. For more information on filters and operators that you can configure, see URI conventions for OData Version 2.0.
Sort	Not applicable.

The following table describes the advanced source properties that you can configure for an SAP OData V2 source:

Property	Description
Page Size	Determines the number of rows to read in each request. Default is 1000.
Number of Rows to Skip	Sets the number of rows you can skip. For example, you can set this property to n to skip the first n rows in a source and read from the (n+1) row. Default is 0. If you enter the number of rows to skip, do not enter the $skip query in the **Query Data Options** field in the advanced source properties.
Number of Rows to Fetch	Determines the number of rows you can read from the source. For example, specify a value of 100 to read 100 records from the source. Specify 0 to read all records from the source. Default is 0. If you enter the number of rows to read, do not enter the $top query in the **Query Data Options** field in the advanced source properties.
Maximum Retry Attempts	The maximum number of retry attempts made to receive the response from the OData V2 endpoint. Default is 1.
Retry Interval in seconds	The time in seconds to wait between each retry attempt. Default is 5.

Property	Description

Query Data Options	Additional data queries to use when you read data from the SAP OData V2 sources.
	You can enter multiple data queries, separated by an ampersand (&), in the following format:
	`<Query option1>=<value>&<Query option2>=<value>&<Custom query3>=<value>....`
	For example, if you want to filter records with a unique identifier greater than or equal to 4500000000000190001 and sort them in the descending order by their unique identifiers, enter the following data queries:
	`$filter=Uniqueid ge '450000000000190001'&$orderby=Uniqueid desc`
	When you add data queries, ensure that there is no space before and after the equal sign (=).
	If you enter the $filter, $top, or $skip query in this field, do not configure a filter condition in the **Filter** field, or enter values in the **Number of Rows to Fetch** or **Number of Rows to Skip** field, respectively.
	For more information about the list of data queries that you can configure, see Query options in the SAP documentation.

Enable Server-Side Pagination	Select to fetch data in smaller chunks by enabling pagination on the SAP server side when you read data from the SAP OData V2 sources. Default is disabled. Before you enable this property, ensure that the SAP OData service is enabled for the server-side pagination using the `$skiptoken` query option in SAP. For more information about how to use the `$skiptoken` query option in SAP, see Implement $skiptoken query option in the SAP documentation.
Tracing Level	Determines the amount of detail that appears in the log file. Select one of the following options from the list: - Terse - Normal - Verbose Initialization - Verbose Data Default is Normal.
Cache Size for Web Service Response (KB)	The memory in KB available for the web service response. If the source contains many rows or columns, you might want to increase the cache size. Default is 1024.
Optimize transformation	Select to improve performance for processing records at run time. Default is disabled.

SAP OData V2 targets in mappings

To write data to entities that are exposed and accessible through OData V2-compliant applications in SAP, configure an SAP OData V2 object as a target in a mapping.

Specify the name and description of the SAP OData V2 target. Configure the target and advanced properties for the target object.

The following table describes the target properties that you can configure for an SAP OData V2 target:

Property	Description
Connection	Name of the target connection. You can select an existing connection or create a new connection for the target connection property.
Target Type	Type of the target object. Select **Single Object** as the target type. Default is Single Object.
Object	Name of the SAP OData V2 target object that you want to create. Select the target object for a single target. You can also select an existing object from the list.
Operation	Type of the target operation. Select one of the following operations from the list: - Insert - Update - Upsert (Update or Insert) - Delete - Data Driven
Update Columns	Select the unique key column as the condition field for the update, upsert, delete, or data driven operation in an SAP OData V2 target.
Data Driven Condition	Enables you to define expressions that flag rows for an insert, update, reject, or delete operation. This property appears only when the operation type is **Data Driven**. You can use one of the following constants in an expression: - DD_INSERT - DD_UPDATE - DD_REJECT - DD_DELETE

The following table describes the advanced target properties that you can configure for an SAP OData V2 target:

Property	Description
Forward Rejected Rows	Determines whether the transformation passes rejected rows to the next transformation or drops rejected rows. By default, the mapping task forwards rejected rows to the next transformation. If you select the **Forward Rejected Rows** option, the agent flags the rows for reject and writes them to the reject file. If you do not select the **Forward Rejected Rows** option, the agent drops the rejected rows and writes them to the session log file. The agent does not write the rejected rows to the reject file.
Cache Size for Web Service Request (KB)	The memory in KB available for the web service request. If the target contains many rows or columns, you might want to increase the cache size. Default is 1024.
Transaction Commit Control	Not applicable.

Property	Description
Transformation Scope	The method in which the mapping applies the transformation logic to incoming data. Select one of the following transformation scopes from the list: - All Input. Processes data from a source that is not real-time. - Transaction. Processes data from a real-time source. Default is All Input. You can connect only one effective real-time source to a real-time target.
Optimize transformation	Select to improve performance for processing records at run time. Default is disabled.

Rules and guidelines for SAP OData V2 objects

Consider the following rules and guidelines for SAP OData V2 objects used as sources and targets in mappings and mapping tasks:

General guidelines

Consider the following general guidelines when you configure a mapping or task:

- A mapping might fail with a data truncation error when you read a large data set. To resolve this error, reduce the page size in the advanced source properties.
- A mapping fails when you use a Joiner transformation to join data from multiple sources that use different authentication types to connect to an SAP OData V2 service.
- You can't display field names by labels when you configure source or target fields in a mapping.
- Data corruption occurs in the target if you specify the $select query in the **Query Data Options** field for SAP OData V2 sources that contain hierarchical entities.

Filter operations

Consider the following rules and guidelines when you filter data from the source in a mapping or task:

- You can't configure data filters on child objects for hierarchical entities.
- To use advanced data filters when you filter data from an SAP OData V2 source, ensure that the filter condition contains only the field name, operator, and field value. You need to remove any additional text that is appended to the filter condition.
- You can't use a parameter in a filter expression in a simple data filter when you filter data from an object.
- You can use only the following filter operators in a simple filter:

  Equals

  Not Equals

  Less Than

  Less or Equals

  Greater Than

  Greater or Equals

Data types

To read or write data of the Edm.Time, Edm.DateTime, or Edm.DateTimeOffset data type, ensure that the source or target field contains the corresponding transformation data type based on the source or target object that you use in a mapping.

Parameterization

Consider the following rules and guidelines when you parameterize the connection in a mapping or task:

- You can't parameterize an SAP OData V2 connection used in the Source or Target transformation.
- You can't overwrite the source and target connection properties at run time.

Object metadata

Entities that have a cyclic relationship with each other are ignored and the task does not fetch the metadata for such entities.

For example, if a business partner has a relationship with contact details and contact details in turn have a relationship with business partner, then the business partner details within contact fields are ignored and not displayed.

Partitioning

You can't configure partitioning for the SAP OData V2 source.

Data preview

You can't preview data of an SAP OData V2 object that contains hierarchical data.

CHAPTER 4

Data type reference

Data Integration uses the following data types in mappings, dynamic mapping tasks, and mapping tasks with SAP OData V2:

Native data types

SAP OData V2 data types appear in the Source and Target transformations when you choose to edit metadata for the fields.

Transformation data types

Set of data types that appear in the transformation. They are internal data types based on ANSI SQL-92 generic data types, which Data Integration uses to move data across platforms. Transformation data types appear in all remaining transformations in a mapping, or mapping task.

When Data Integration reads source data, it converts the native data types to the comparable transformation data types before transforming data. When Data Integration writes to a target, it converts the transformation data types to the comparable native data types.

Flat SAP OData V2 and transformation data types

Use only the compatible flat SAP OData V2 data types in mappings, dynamic mapping tasks, and mapping tasks to read from or write to OData V2-compliant applications in SAP.

The following table lists the flat SAP OData V2 data types that you can use with SAP OData V2 Connector, along with the corresponding transformation data types when you read from or write to OData V2-compliant applications in SAP:

Flat SAP OData V2 Data Type	Transformation Data Type When You Read	Transformation Data Type When You Write
Edm.Binary	Binary	Binary
Edm.Boolean	String	String
Edm.Byte	Integer	Integer
Edm.DateTime	Date/Time	Date/Time
Edm.DateTimeOffset	Date/Time	Date/Time
Edm.Double	Double	Double

Flat SAP OData V2 Data Type	Transformation Data Type When You Read	Transformation Data Type When You Write
Edm.Decimal	Decimal	Decimal
Edm.Int16	Integer	Integer
Edm.Int32	Integer	Integer
Edm.Int64	BigInt	BigInt
Edm.SByte	Integer	Integer
Edm.Single	Double	Double
Edm.String	String	String
Edm.Time	Date/Time	Date/Time

Hierarchical SAP OData V2 and transformation data types

Use only the supported hierarchical SAP OData V2 data types in mappings, dynamic mapping tasks, and mapping tasks to read from OData V2-compliant applications in SAP.

The following table lists the hierarchical SAP OData V2 data types that you can use with SAP OData V2 Connector, along with the corresponding transformation data types when you read from OData V2-compliant applications in SAP:

Hierarchical SAP OData V2 Data Type	Transformation Data Type
Edm.Binary	String
Edm.Boolean	String
Edm.Byte	Integer
Edm.DateTime	String
Edm.DateTimeOffset	String
Edm.Double	Double
Edm.Decimal	Decimal
Edm.Int16	Integer
Edm.Int32	Integer
Edm.Int64	Long

Hierarchical SAP OData V2 Data Type	Transformation Data Type
Edm.SByte	Integer
Edm.Single	Double
Edm.String	String
Edm.Time	String

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